Miguel Alves

Curriculum Vitae

Departamento de Matemática
Escola de Ciências da Universidade do Minho
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Information

Full Name: Miguel José de Melo Alves.

Date of Birth: 2nd January 2001.

Nationality: Portuguese.

Research Interests

Logic, Type Theory, λ-calculus, Proof Assistants, Automated Theorem Proving, Functional Programming.

Education

2023 – 2025 **MSc in Mathematics and Computation**, *Escola de Ciências da Universidade do Minho, Portugal*, Specialisation in Computation.
Grade: 19/20

2019 – 2022 **BSc in Computer Science**, *Faculdade de Ciências da Universidade do Porto, Portugal*, Complementary Formation Centred in Mathematics.
Grade: 17/20

Publications

Conference and Workshop Papers

Thesis

2025 **M. Alves**, On the mechanisation of the multiary lambda calculus and subsystems, MSc thesis.
Advised by Luís Pinto and José Espírito Santo
Grade: 20/20

Grants

October, 2024 – March, 2025 **Research Fellowship**, *Centro de Matemática da Universidade do Minho*, UIDB/00013/2020.
Escola de Ciências da Universidade do Minho.

January, 2023 – June, 2023 **Research Fellowship**, *Centro de Matemática da Universidade do Porto*, UIDB-00144-2020.
Departamento de Matemática da Faculdade de Ciências da Universidade do Porto.

March, 2022 – December, 2022 **Research Initiation Fellowship**, *Centro de Matemática da Universidade do Porto*, UIDB-00144-2020.
Departamento de Matemática da Faculdade de Ciências da Universidade do Porto.

Activities

Posters

2025 **On the formalisation of the multiary λ-calculus and subsystems**, in *Dia da Pós-graduação e Empregabilidade em Ciências, Braga, Portugal*.

Oral Presentations

2026 **Formalizing Calculi with Unconventional Substitution**, in *LFMTP '26, Lisboa, Portugal*.

2026 **On the mechanisation of the multiary lambda-calculus and subsystems**, in *Days in Logic '26, Braga, Portugal*.

2026 **On the mechanisation of the multiary lambda-calculus and subsystems**, in *Days in Logic '26, Braga, Portugal*.

2025 **On the mechanisation of the multiary lambda-calculus and subsystems**, in *The 2025 Edition of the ALC Meeting, Braga, Portugal*.

2024 **On the mechanisation of a fragment of the simply-typed lambda-calculus and its sequent calculus counterpart**, in *The 2024 Edition of the ALC Meeting, Braga, Portugal*.

Teaching

Spring 2026 **Computational Models**, *CC1004*, in Faculdade de Ciências da Universidade do Porto.

Outside academia

Volunteer work

2025 – 2026 **Monthly volunteer**, *Fonte da Prata*, Portugal.

2022 – 2024 **Summer camp monitor**, *Fonte da Prata*, Portugal.

2020 – 2023 **Youth group animator**, *CREU-IL*, Porto, Portugal.

Teaching experience

2025 – 2026 **Mathematics teacher**, *Colégio Luso-Frances*, Porto, Portugal.

2020 – 2025 **One-to-one tutoring in Mathematics**.

Musical activity

2020 – present **Pomadinha music project**, drummer and composer.

2010 – 2019 **Secundary education in Percussion**, *Escola de Música de Perosinho*, Portugal.